Section A: Structured Questions

Answer **all** the questions in the spaces provided.

1 **(a)** Fig. 1.1 shows the second ionisation energies of eight elements with consecutive proton numbers and one of the elements is aluminium.

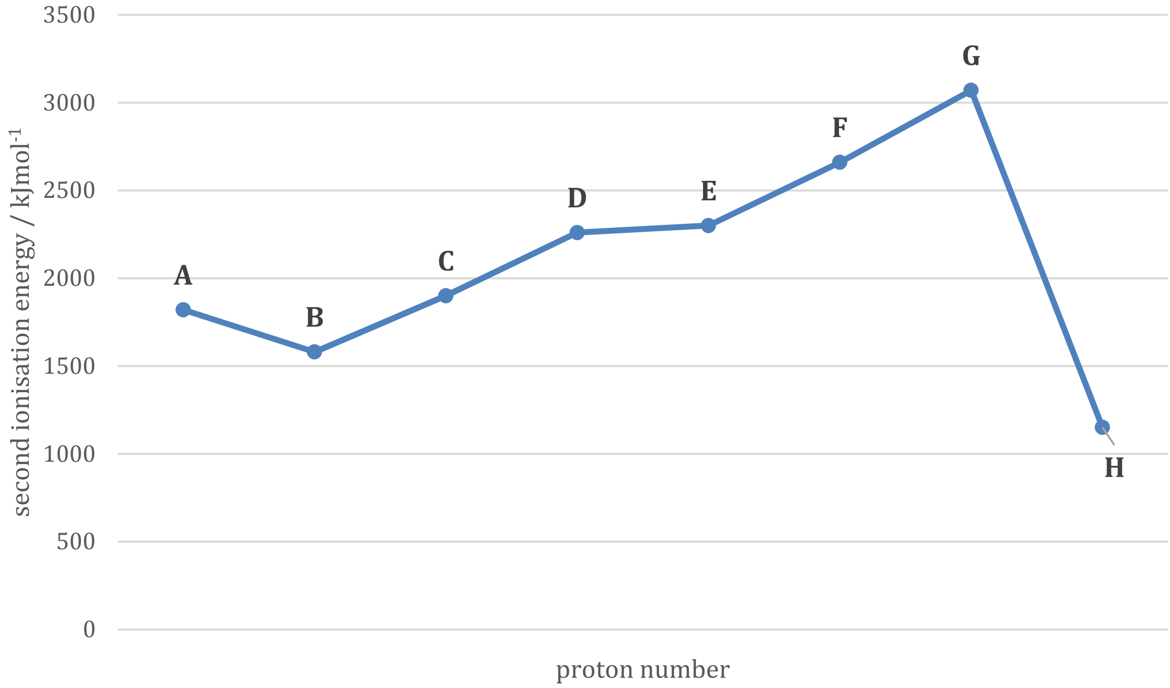


Fig.1.1

(i) Define, with the aid of an equation, what is meant by the second ionisation energy of aluminium.

$Al^{+}(g) \rightarrow Al^{2+}(g) + e$

Second ionisation energy is the energy required to remove one mole of outermost electrons from one mole of Al^{+}(g) cations to form one mole of Al^{2+}(g) cations.

[2]

(ii) Explain why the second ionisation energy of aluminium is usually more endothermic than the first ionisation energy.

1st IE: $Al(g) \rightarrow Al^{+}(g) + e$

2nd IE: $Al^{+}(g) \rightarrow Al^{2+}(g) + e$

The general increase is due the increasing positive charge on the cation.
As successive electrons are being removed, the same number of protons is attracting fewer electrons.

Consequently, there is an increase in the effective nuclear charge of the cation and requires more energy to remove the next electron.

[2]

(iii) Identify the element which represents aluminium. Explain the reasoning.

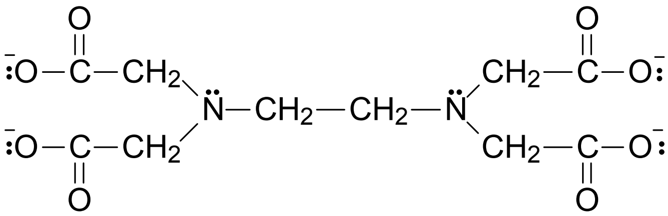
The sharp drop in 2nd IE value from **G** to **H** indicates the removal of the second electron from the inner shell, thus **G** is an element in Group 1. **[any other logical deduction]**

Element **A** is in Group 13 and is aluminium.

[2]

(b) Hard water is high in dissolved minerals, specifically calcium ions. Although hard water does not pose a health risk, it causes poor soap performance. The soap used in hard water combines with the dissolved minerals to form a sticky soap curd which consists of insoluble salts.

(i) Ethylenediaminetetraacetate or EDTA is a component in laundry detergent.



EDTA

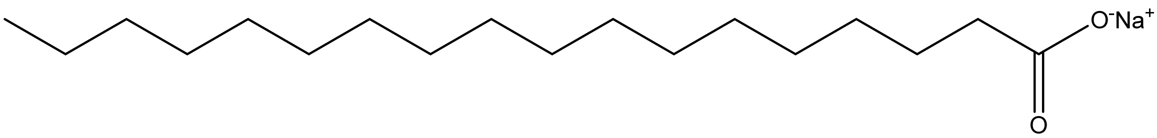
Suggest how EDTA could improve the soap performance in hard water.

EDTA forms a complex with metal ion Ca^{2+}, thus removing Ca^{2+}.

[1]

(ii) Soap has effective cleaning properties because it contains one or more surfactants.

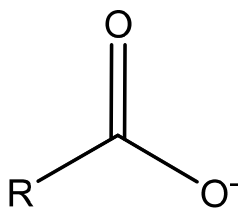
A surfactant is a molecule with a *non-polar* tail and a *polar* head. An example is sodium stearate, as shown below.



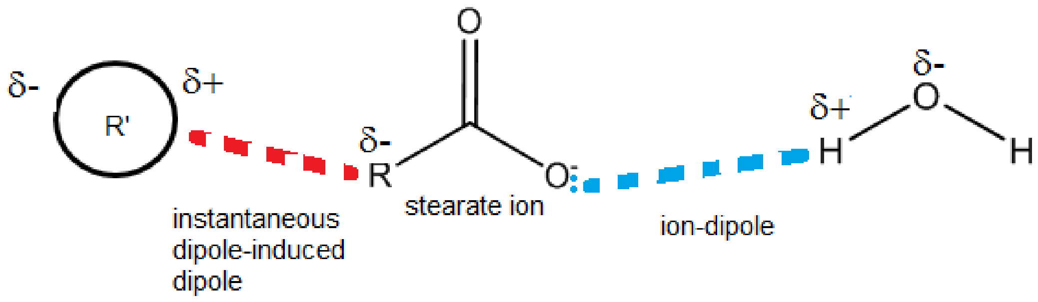
A simplified representation of stearate ion is $RCOO^-$.

Complete the diagram by drawing the relevant dipoles and indicate the type of intermolecular forces of attraction between stearate ion, an oil molecule and a water molecule. Hence explain how stearate ion helps to remove oil from surface during washing.

You may use R' to represent an oil molecule.



stearate ion



The surfactant/ stearate ion is able to interact with both water and oil - the negatively charged COO^- of the surfactant forms ion-dipole interactions with water, and the non-polar tail forms instantaneous dipole-induced dipole (id-id) interactions with oil (non-polar). Thus it can help to remove oil.

[3]

[Total: 10]

2 The table below shows the melting points of some of the chlorides of the elements in period 3 of the periodic table.

Chlorides	Melting points/ °C
Magnesium chloride, , $MgCl_2$	714
Aluminium chloride, $AlCl_3$	192
Phosphorus pentachloride, PCl_5	161
Sulfur dichloride, SCl_2	–121

(a) Account, in terms of structure and bonding, the difference in the melting points between

(i) magnesium chloride and aluminium chloride

$MgCl_2$ has a giant ionic structure while $AlCl_3$ has a simple molecular structure.

More energy is required to overcome the electrostatic forces of attraction between Mg^{2+} and Cl^- compared to instantaneous dipole-induced dipole attraction between $AlCl_3$ molecules. Hence $MgCl_2$ should have a higher melting point.

[2]

(ii) phosphorus pentachloride and sulfur dichloride

Both PCl_5 and SCl_2 have simple molecular structure.

PCl_5 has larger and more polarisable electron cloud than SCl_2. More energy is required to overcome the stronger instantaneous dipole-induced dipole attraction between molecules of PCl_5 compared to molecules of SCl_2. Hence PCl_5 should have a higher melting point.

[2]

(b) When PCl_5 is added to water, liquid phosphorus oxychloride, $POCl_3$ is formed as an intermediate compound.

(i) Reactions of covalent chlorides with water can be rationalised as step–wise replacement of –C*l* with –OH. Deduce a three–step reaction sequence for the formation of $POCl_3$ from PCl_5.

Step 1: $PCl_5 + H_2O \rightarrow PCl_4OH + HCl$

Step 2: $PCl_4OH + H_2O \rightarrow PCl_3(OH)_2 + HCl$

Step 3: $PCl_3(OH)_2 \rightarrow POCl_3 + H_2O$

[3]

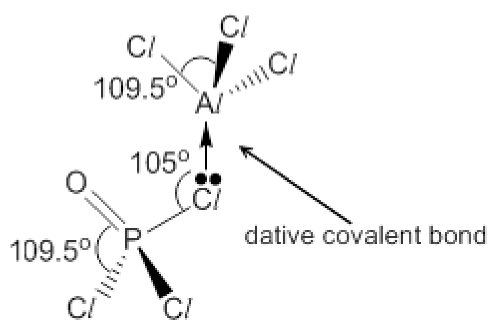
$POCl_3$ can also acts as a *Lewis base* to remove the $AlCl_3$ at the end of a Friedel–Crafts reaction, resulting in the formation of a covalent product.

(ii) Define the term '*Lewis base*'.

A Lewis base is an electron–pair donor.

[1]

(iii) Draw a diagram to illustrate the shape of the product which results in the removal of $AlCl_3$, stating the type of bonding present and the bond angles around the central atoms.



Type of bonding: **Dative covalent bond , (accept if student draw O to Al.)**

[2]

[Total: 10]

3 **(a)** When steam is passed over hot coke (a form of carbon obtained from coal), water gas is produced. Water gas, consists of a mixture of carbon monoxide and hydrogen, is an important industrial fuel.

$$C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$$

When 0.100 mol of steam was reacted with coke in a vessel of 1.00 dm^3 and allowed to reach equilibrium at 800 °C, the partial pressure of steam was found to be 2.67 atm.

(i) Calculate the initial pressure of steam, in atm, introduced into the vessel.

Using pV = nRT

$p = \frac{(0.1 \times 8.31 \times (800 + 273))}{(1 \times 10^{-3})} = 8.9166 \times 10^5$ Pa = 8.80 atm (1 atm = 101325 Pa)

[2]

(ii) Using your answer in **a(i)**, calculate a value for K_p, in atm, at 800 °C.

	C(s)	$H_2O(g)$	CO(g)	$H_2(g)$
Initial P/atm	-	8.80	0	0
Change/atm	-	- 6.13	+6.13	+6.13
Equilibrium P/atm	-	2.67	6.13	6.13

$K_p = \frac{p_{CO} p_{H_2}}{p_{H_2O}} = \frac{(6.13)^2}{2.67} = 14.1\ atm$

[2]

(iii) Calculate the minimum mass of carbon required in the vessel.

Since 6.13 atm of steam was reacted,

number of moles of steam = $\frac{pV}{RT} = \frac{(6.13 \times 101325 \times 10^{-3})}{8.31 \times (800+273)} = 0.06966 mol$

Mass of C = 0.06966 × 12.0 = 0.836 g

[2]

(iv) State and explain whether high or low pressure would favour a higher yield of water gas.

To obtain higher yield, the position of equilibrium must shift to the right.
Since the forward reaction produces greater number of moles of gaseous molecules, low pressure favours higher yield.

[2]

(v) The reaction was repeated at a lower temperature and the numerical value of K_p was found to have decreased.

State and explain whether the forward reaction is endothermic or exothermic.

At low temperature, K_p value decreased. This shows that the equilibrium position shifts to the left which produces heat (exothermic).
Hence forward reaction is endothermic.

[2]

(b) **(i)** The relationship between $\Delta G^{\ominus}$ and K_p for the reaction between carbon and steam is

$$\Delta G^{\ominus} = -RT \ln K_p$$

where $\Delta G^{\ominus}$ is the standard Gibbs free energy change in J mol^{-1}, R is the gas constant, T is the temperature in Kelvin at which the equilibrium is established and K_p is the equilibrium constant.

Given that the K_p of the reaction at 298K is 2.97 x 10^{-18} atm, calculate the $\Delta G^{\ominus}$ of the reaction under standard conditions.

$$\Delta G^{\ominus} = -8.31\ (298) \ln (2.97 \times 10^{-18} \times 101325)$$
$$= +7.14 \times 10^{4} \text{ J mol}^{-1}$$

[1]

(ii) Comment on the sign of $\Delta G^{\ominus}$ with reference to the position of equilibrium of the reaction under standard condition.

The sign of $\Delta G^{\ominus}$ is positive, the reaction is not spontaneous under standard conditions and position of equilibrium lies to the left.

[2]

[Total: 13]

4. **(a)** Ethers have the general formula, R-O-R' (where R and R' are alkyl or aryl groups).

The most useful method of preparing ethers is by the Williamson ether synthesis, in which an alkoxide ion (RO^-) reacts with an alkyl halide (R'X) in an S_N2 mechanism.

A reaction scheme is shown below:

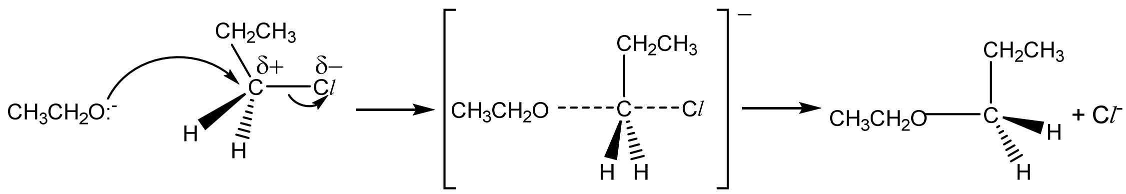
$$ROH \xrightarrow[\text{step } \mathbf{1}]{NaH} RO^-Na^+ \xrightarrow[\text{step } \mathbf{2}]{R'X \text{ (alkyl halide)}} ROR'$$

(i) Suggest why step **1** is necessary in the Williamson ether synthesis.

To generate a stronger nucleophile

[1]

(ii) Using 1-chloropropane as the alkyl halide and $CH_3CH_2O^-$ as the alkoxide ion, outline the mechanism for **step 2** of the Williamson ether synthesis. Show all charges and relevant lone pairs of electrons and show the movement of electron pairs by using curly arrows.



[3]

(iii) The same reaction in **a(ii)** was repeated using iodopropane instead of chloropropane. Using relevant data from the *Data Booklet*, state and explain the effect on the rate of reaction.

The rate of reaction using iodopropane will be faster.
From the Data Booklet,
Bond Energy (C-C*l*) = 340 $kJmol^{-1}$;
Bond energy (C-I) = 240 $kJmol^{-1}$.
The C-I bond is weaker, thus less energy is needed to break the bond.

[2]

(iv) Suggest why the product obtained in **a(ii)** shows no optical activity.

There is absence of chiral carbon, thus it is unable to form enantiomers.

[1]

Fig. 4.1 shows two possible reaction sequence to synthesise compound **P**.

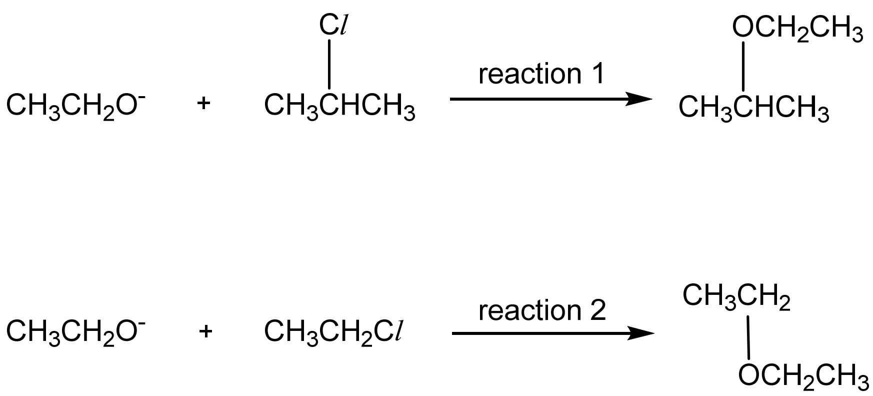


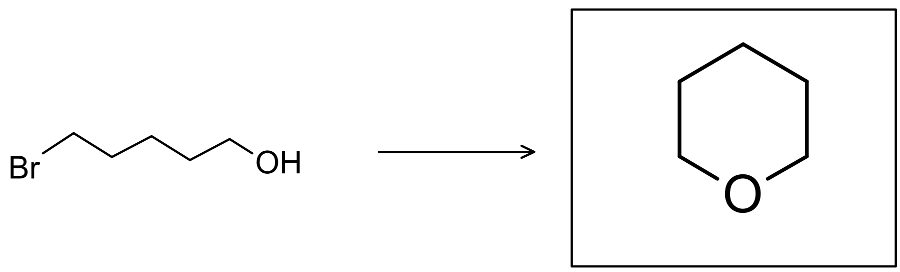
Fig 4.1

(v) State and explain which reaction will give a higher yield.
Reaction 2 will give higher yield.

The Williamson ether synthesis method proceeds via S_N2 mechanism which is favoured by primary halogenoalkane as there is less steric hindrance posed to the attacking nucleophile.

[2]

(vi) Suggest the structure of the product when 5-bromo-pentan-1-ol undergoes the Williamson ether reaction.



5-bromopentan-1-ol

[1]

(b) Halogenated organic compounds can be synthesised from various functional groups.

Chlorofluorocarbons (CFCs) have been widely used in aerosol sprays, refrigerants, but are now known to destroy ozone in the upper atmosphere through a free radical chain reaction.

(i) A student wrongly proposed that a propagation step for the formation of 1–chlorobutane from butane to be:

$$CH_3CH_2CH_2CH_3 + Cl\bullet \rightarrow CH_3CH_2CH_2CH_2Cl + H\bullet$$

Write the correct propagation step. Hence use relevant data from the *Data Booklet*, calculate the enthalpy change of reaction to explain why the propagation step proposed by the student is incorrect.

The correct propagation step is:
$CH_3CH_2CH_2CH_3 + Cl\bullet \rightarrow CH_3CH_2CH_2\dot{C}H_2 + HCl$

BE (C – H) = 410 kJ mol^{-1} BE (H – C*l*) = 431 kJ mol^{-1} ; BE (C –C*l*) = 340 kJ mol^{-1}

ΔH_r for the incorrect propagation step is 410 – 340 = = +70 kJ mol^{-1}
ΔH_r for the correct propagation step is 410 – 431 = –21 kJ mol^{-1}

Since the ΔH_r for the correct propagation step is more exothermic, the reaction is more spontaneous.

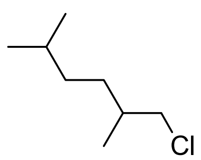
[3]

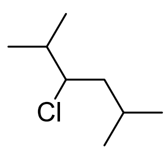
(ii) 2,5-dimethylhexane is reacted with chlorine in the presence of UV light to form 3 monochlorinated alkanes.

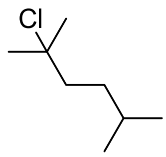
It has been found experimentally that in free radical substitution of alkanes, the primary, secondary and tertiary hydrogen atoms are replaced by chlorine atoms at different rates, as shown in the following table.

type of hydrogen atom	**reaction**	**relative rate**
primary	$RCH_3 \rightarrow RCH_2Cl$	1
secondary	$R_2CH_2 \rightarrow R_2CHCl$	7
tertiary	$R_3CH \rightarrow R_3CCl$	21

Draw the structural formula of the 3 possible monochlorinated products of 2,5-dimethylhexane and using the information in the table, suggest the relative ratios in which they are formed.





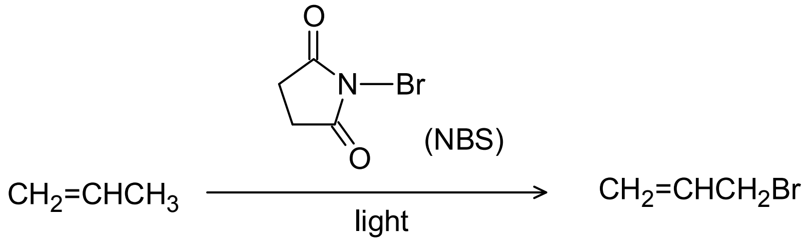


Ratio (12 × 1) = 12 (4×7) = 28 (2×21) = 42
6 14 21

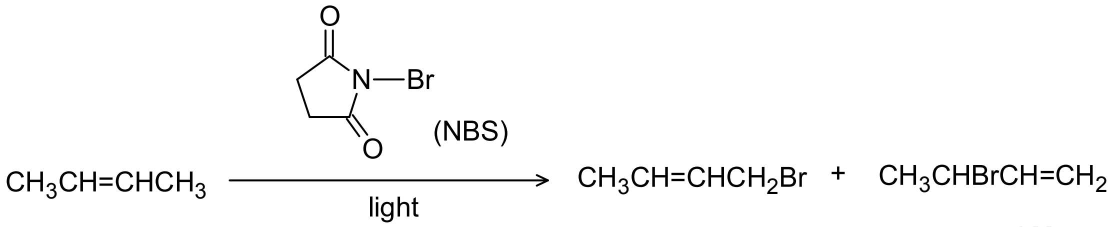
[3]

(c) Another method of preparing alkyl halides from alkenes is by the reaction with N-bromosuccinimide (NBS) in the presence of light to give products which are substituted at the allylic position i.e. the position *next* to the double bond. NBS is used as a source of bromine.

The process is similar to the free radical substitution mechanism of alkanes.



(i) However, a mixture of 2 products (**W** and **X**) is formed when but-2-ene undergoes this reaction.



compound **W** compound **X**

Suggest a reason for the formation of compound **X**.

Another resonance structure can be obtained due to the delocalisation of the electron.

secondary radical is more stable than primary radical as it has more electron-donating group or
$CH_3CH{=}CH\dot{C}H_2 \leftrightarrow CH_3\dot{C}H{-}CH{=}CH_2$

[2]

(ii) State the type of isomerism shown by compound **W** and compound **X.**

They are constitutional / positional isomers.

[1]

[Total: 19]

5 Infra-red(IR) absorptions can be used to identify functional groups in organic compounds. For example, ethyl ethanoate shows absorption at 100 – 1300 cm^{-1} and 1680 – 1750 cm^{-1}.

A student used infra-red spectrometer to obtain an infra-red(IR) spectrum of benzyl alcohol $C_6H_5CH_2OH$ as shown in Fig. 5.1.

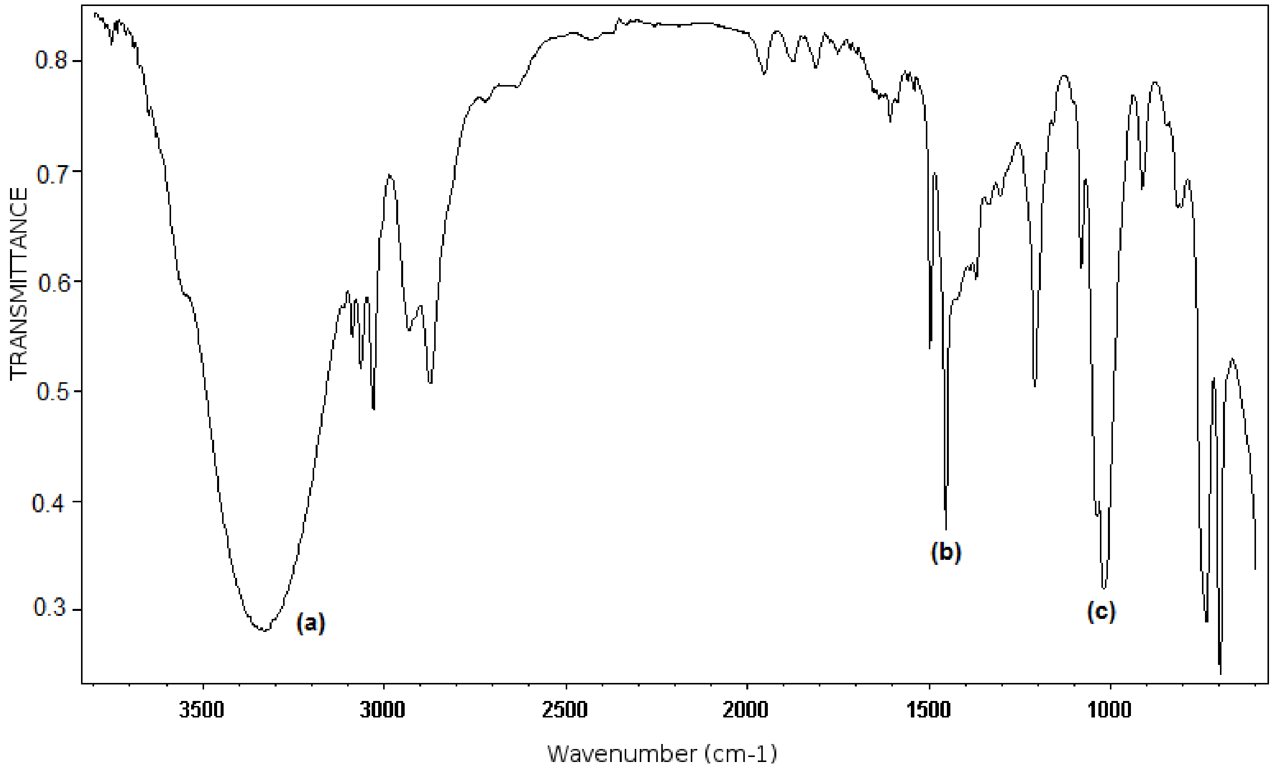


Fig. 5.1

(a) **Use the table of characteristic infra-red absorption frequencies in the *Data Booklet* to** identify the infra-red absorption range shown by benzyl alcohol.

IR region	Wavenumbers/ cm^{-1}	Functional groups present
(a)	3200 – 3600	(Primary) alcohol
(b)	1475 – 1625	Benzene
(c)	970 – 1260	(Primary) alcohol

[3]

(b) The student carried out reflux of benzyl alcohol with acidified potassium dichromate and the product **A** obtained was analysed using spectrometer. Fig. 5.2 shows the IR spectrum obtained.

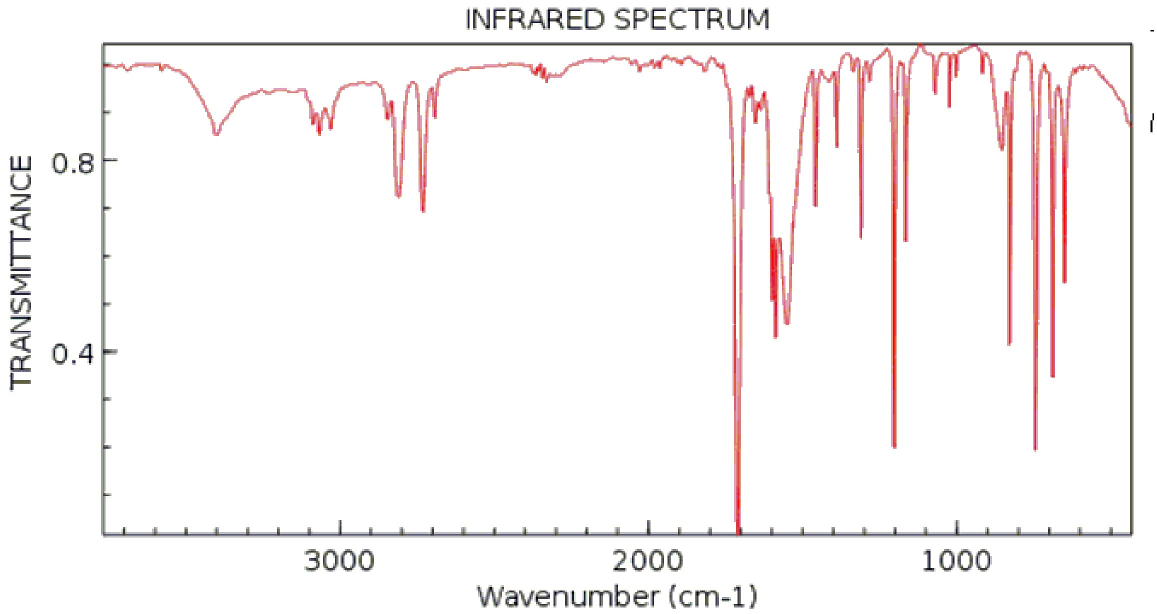
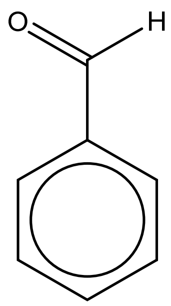


Fig. 5.2

By comparing the two infra-red spectra, suggest two differences from the infra-red spectra and draw the structure of the product **A** obtained.

Product **A:**



1. Peak (a) from Fig 5.1 is missing compared to Fig 5.2., indicating loss of –OH group
2. Presence of C=O at 1670–1740 cm^{-1} in Fig 5.2 compared to Fig 5.1, could be due to aldehyde group

Note: RCOOH is not acceptable as there is no O–H absorption from 2500-3000 cm^{-1} in Fig 5.2

[3]

(c) Suggest a simple chemical test that could be used to confirm benzyl alcohol $C_6H_5CH_2OH$ has been completely converted into product **A** in **(b)**.

Test : Add anhydrous $SOCl_2$, PCl_5 or Na(s)

Observations: Absence of HCl white fumes confirmed that there is no alcohol present or absence of effervescence of H_2 gas.

[2]

(d) A student wishes to determine the concentration of product **A** obtained. The student prepares 5 solutions of known concentrations, 0.0100, 0.0200, 0.0400, 0.0800 and 0.100 mol dm^{-3} of product **A**. 10 μL of each solution with the sample was used and a high performance liquid chromatogram was obtained. The peak areas of the chromatographs are given in the table below and a graph of concentration against peak area was obtained in Fig.5.3.

Solution concentration / mol dm^{-3}	0.0100	0.0200	0.0400	0.0800	0.100	sample
Peak area / cm^2	0.50	0.83	1.39	2.63	3.18	2.53

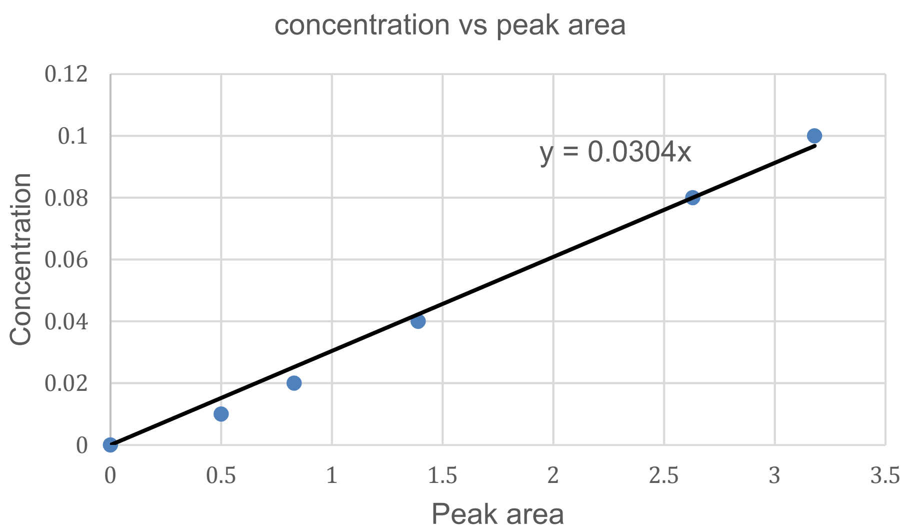


Fig. 5.3

(i) Using the peak area of the sample from the table and information from the graph, determine the concentration of product **A** in the sample.

Concentration of product **A** = 0.0304(2.53) = 0.0769 mol dm^{-3}

[1]

(ii) Suggest what the student can do to ensure high yield of product **A** from the reaction of benzyl alcohol with acidified potassium dichromate in **(b)**. Use Le Chatelier's Principle to explain your suggestion.

$C_6H_5CH_2OH$ + [O] $\rightleftharpoons$ C_6H_5CHO (product **A**) + H_2O
Carry out immediate distillation of product **A**.
As product **A** is being removed, position of equilibrium shifts right to increase the concentration of product **A**, hence yield increases.

[2]

(e) Fig. 5.4 shows a Ritter reaction.

The Ritter reaction transforms a nitrile to amide using a strong acid and water as follows.

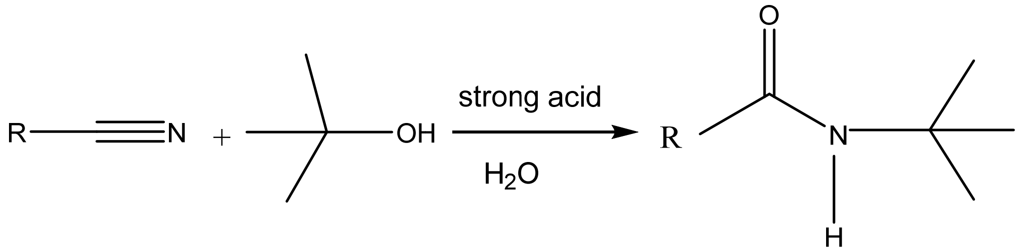
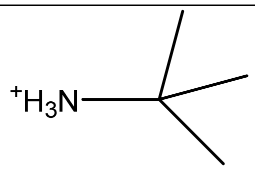


Fig. 5.4

(i) Side products will be formed from the Ritter reaction if heating was carried out.

Suggest the type of reaction which occurs and draw the structures of two side products formed.

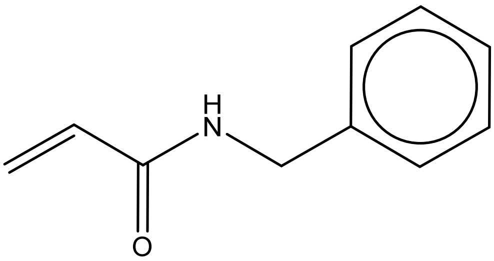
Type of reaction: Hydrolysis



RCOOH

[3]

(ii) Benzyl alcohol, $C_6H_5CH_2OH$ undergoes Ritter reaction with CH_2CHCN as shown in Fig. 5.4. Draw the structure of the amide formed.



(f) Fig. 5.5 shows a possible process in the human body where benzyl alcohol is oxidised to benzoic acid, conjugated with amino acid, glycine in the liver, and excreted as benzoylaminoethanoic acid, as shown.

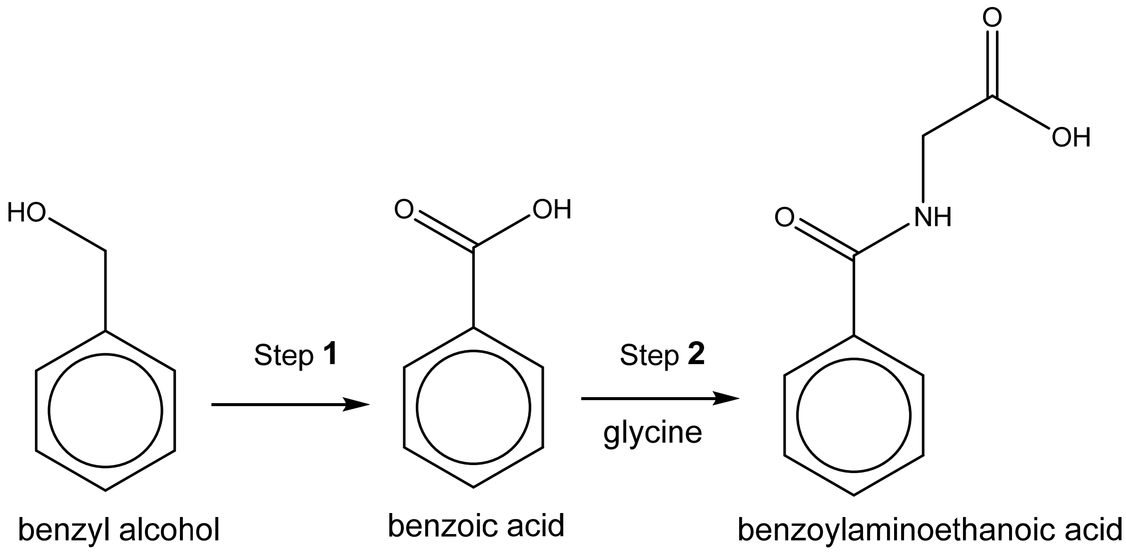


Fig 5.5

(i) Write an equation for the oxidation of benzyl alcohol to form benzoic acid. Use [O] to represent the formula of the oxidising agent.

$C_6H_5CH_2OH + 2[O] \rightarrow C_6H_5COOH + H_2O$

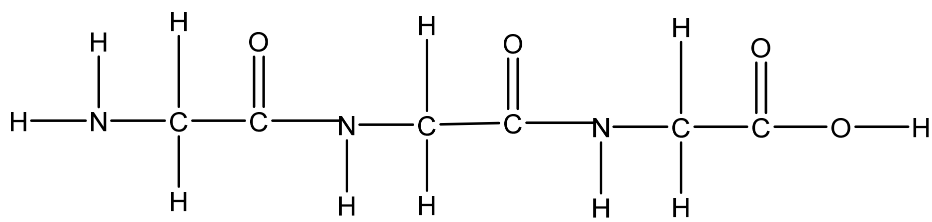
[1]

(ii) What type of reaction takes place in step **2**?

Condensation/Nucleophilic substitution

[1]

(iii) Draw the displayed formula of a tripeptide consisting of glycine only.



[1]

(iv) Benzoylaminoethanoic acid can be reduced by $LiAlH_4$ in dry ether at room temperature and hydrogen gas with nickel catalyst under high heat and pressure. The organic products formed in the two reactions are different. Compound **B** has a molecular mass of 151 and compound **C** has a molecular mass of 185.

Deduce the structures of **B** and **C**.

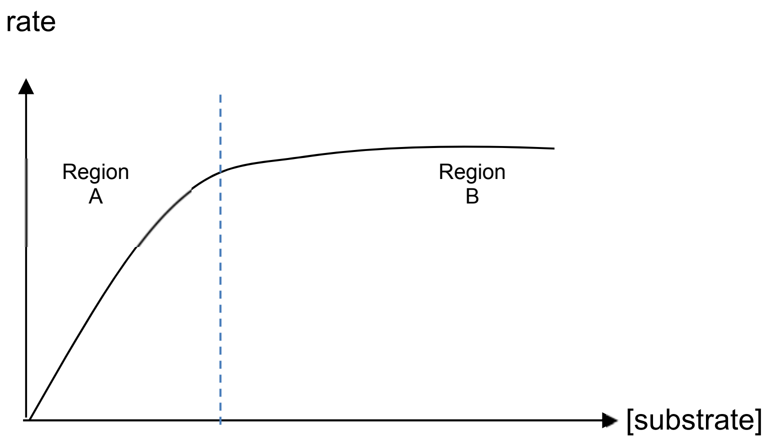
compound **B**

compound **C**

[2]

The synthesis of glycine in the liver is catalysed specifically by glycine synthase enzyme.

(v) Describe,with the aid of a sketch how the rate of synthesis of glycine is affected by the concentration of the substrate.



At low [substrate], the **active sites are not filled** and rate = k [substrate]; i.e. rate is proportional to [substrate] and the reaction is first order wrt [substrate] (region A).

At high [substrate], increasing its concentration has no effect at all on the rate of reaction; as **all the active sites are occupied**. i.e. reaction is zero order with respect to [substrate] (region B).

[3]

[Total: 23]